LESSON 327 (1.6 DUAL XC, 1.4 INSTRUMENT)

Lesson Objective
During this lesson, the student will plan and conduct a short IFR cross-country flight. During the flight, the student will become more familiar with IFR departure, enroute, and arrival procedures.

Briefing
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Lesson Content (Ground)
Go/no-go Decision
- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)
- Emphasize the importance of proper setup prior to taxi and takeoff; the flight plan should be loaded in the GPS, the departure heading and altitude should be bugged on the PFD, and the departure frequency and transponder code should be set prior to entering the movement area.

Calculating ETEs and ETAs
- Ensure the student understands how to calculate ETEs (can be accomplished during flight planning) as well as ETAs (based on actual takeoff time).

Enroute Course Changes
- Ensure the student complies with any course or route changes issued by ATC.

Use of DPs and STARs
- Review any published SIDs/ODPs and consider aircraft performance to ensure they can be complied with.
- KSFB - Sanford usually assigns the Sanford One Departure, a simple radar vector-based SID. This should be briefed on the ground before the flight.

Precision and/or Non-Precision Approaches

- Encourage the student to complete the WIRE checklist as early as practical to reduce the workload upon arrival.
- Ensure the student completes all the steps of the WIRE checklist:
    - **W**eather
    - **I**nstruments:
        - Set both altimeters.
        - Set the correct nav source on the PFD (VLOC/GPS)
        - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
    - **R**adios:
        - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
        - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
        - Set the final approach course.
    - **E**nroute brief

<u>Post-Brief</u>

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 328
- Assign Homework:
    - Plan the next XC. Review the requirements to pick suitable airports.

<u>Completion Standards</u>

At the completion of this flight, the student shall:

1. Be able to explain the departure and arrival procedures that may be encountered on an IFR flight.
2. Know the methods used to calculate ETA's and comply with course changes that may be issued by ATC or necessitated by enroute weather.